Summary in Graph

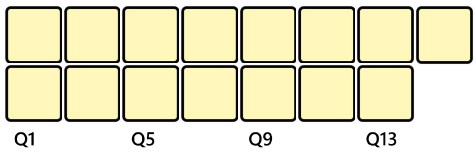
Exam Summary (GO Classes CS/DA Test Series 2026 | Probability and Statistics | Bayes Theorem, Conditional Probability | Topic Wise Test)

Qs. Attempted:	**0** (0 + 0)	**Correct Marks:**	**0** (0 + 0)
Correct Attempts:	**0** (0 + 0)	**Penalty Marks:**	**0** (0 + 0)
Incorrect Attempts:	**0** (0 + 0)	**Resultant Marks:**	**0** (0 + 0)

Total Questions:	**15** (5 + 10)
Total Marks:	**25** (5 + 20)
Exam Duration:	**45 Minutes**
Time Taken:	**0 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (162) out of 1620 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

A gambler has in his pocket a fair coin and a two-headed coin. He selects one of the coins at random and flips it. If the coin comes up heads, what is the probability that it is the fair coin?

A. $\frac{1}{2}$

B. $\frac{1}{3}$

C. $\frac{2}{3}$

D. $\frac{3}{4}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 05sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

In a probabilistic model, it is given that $\mathrm{P}[A] = \mathrm{a}, \mathrm{P}[B] = \mathrm{b}$, and $\mathrm{P}\left[(B^c \mid A^c)\right] = \mathrm{e}$. Determine $\mathrm{P}[B \mid A]$ in terms of a, b, and e.

A. $\mathrm{P}[B \mid A] = \frac{\mathrm{b}-(1-\mathrm{e})(1-\mathrm{a})}{\mathrm{a}}$
B. $\mathrm{P}[B \mid A] = \frac{\mathrm{b}+(1-\mathrm{e})(1-\mathrm{a})}{\mathrm{a}}$
C. $\mathrm{P}[B \mid A] = \frac{\mathrm{b}+(1-\mathrm{e})(1+\mathrm{a})}{\mathrm{a}}$
D. $\mathrm{P}[B \mid A] = \frac{\mathrm{b}-(1+\mathrm{e})(1-\mathrm{a})}{\mathrm{a}}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Mark all expressions which indicate that X is independent of Y given Z.

A. $P(X, Y \mid Z) = P(X \mid Z)P(Y \mid Z)$
B. $P(X, Y, Z) = P(X, Z)P(Y)$
C. $P(X \mid Y, Z) = P(X \mid Z)$
D. None of the above

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Select all expressions that are equivalent to $P(A, B \mid C)$,
given that $A \perp B \mid C$,
where $\perp$ denotes conditional independence — that is,
A is independent of B when C is known.

A. $\frac{P(A|C)P(C|B)P(B)}{P(B,C)}$

B. $\frac{P(A|C)P(B,C)}{P(C)}$

C. $P(A \mid B, C)P(B \mid C)$

D. $\frac{P(C|A,B)P(B|A)P(A)}{P(B|C)P(C)}$

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Probability

Select all expressions that are equivalent to $\mathbf{P}(\mathbf{A} \mid \mathbf{B}, \mathbf{C})$, given that A $\perp$ B | C

A. $\frac{P(C|A)P(A|B)P(B)}{P(C)}$

B. $\frac{P(B,C|A)P(A)}{P(B,C)}$

C. $\frac{P(A|C)P(C|B)P(B)}{P(B,C)}$

D. $\frac{P(C|A,B)P(B|A)P(A)}{P(B|C)P(C)}$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose that the probability of making a mistake on your homework depends on how much sleep you got the previous night. The probabilities are as follows:

- If you got at least 8 hours of sleep, there is a 10% chance of making a mistake.
- If you got between 6 and 8 hours of sleep, there is a 40% chance of making a mistake.
- If you got less than 6 hours of sleep, there is a 75% chance of making a mistake.

Additionally, the probabilities for the amount of sleep are:

- There is a 20% chance that you got at least 8 hours of sleep.
- There is a 50% chance that you got between 6 and 8 hours of sleep.

What is the overall probability that you make a mistake on your homework?

A. 0.265

B. 0.365

C. 0.445

D. 0.545

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A gambler has in his pocket a fair coin and a two-headed coin. He selects one of the coins at random, and then he flips it. What is the probability that the flip results in heads?

A. 0.5

B. 0.75

C. 0.25

D. 1.0

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

In a game where the probability of winning on a single toss of a dice is p, Player A starts and if he fails, he passes the dice to Player B. They continue tossing the dice alternately until one of them wins. What is the probability that Player A wins?

A. $\frac{1}{p}$

B. $\frac{1}{1-(1-p)^2}$

C. $\frac{1}{2-p}$

D. $\frac{p}{2}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider the following statements, where $P(\cdot)$ denotes probability and assuming all conditional probabilities involved are non-zero:
S1: If $P(a \mid b, c) = P(b \mid a, c)$, then $P(a \mid c) = P(b \mid c)$.
S2: If $P(a \mid b) = P(a)$, then $P(b \mid c) = P(b)$.
Choose the correct option that correctly interprets the implications of these statements:

A. Statement S1 is true, and statement S2 is false.
B. Statement S1 is false, and statement S2 is true.
C. Both statement S1 and statement S2 are true.
D. Both statement S1 and statement S2 are false.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Which of the following statements is/are TRUE?

A. For any three events A, B, and C where A and B are disjoint and $0 < P(C) < 1$, $P(A \cup B \mid C) = P(A \mid C) + P(B \mid C)$
B. For any two events A and B where $0 < P(B) < 1$, $P(A \mid B) = 1 - P(A \mid B^c)$
C. For any three events A, B, and C, $P(A \cap B \cap C) = P(A)P(B)P(C)$
D. For any two events A and B, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Given events A_1, A_2, A_3 in a sample space S with probabilities $P(A_1) = 0.4, P(A_2) = 0.2, P(A_3) = 0.4$, and assuming $A_i \cap A_j = \emptyset$ for $i \neq j$, and another event B with conditional probabilities $P(B \mid A_1) = 0.1, P(B \mid A_2) = 0.2, P(B \mid A_3) = 0.05$, what is the probability that either A_1 or A_2 occurs

given that B has occurred?

A. $P(A_1 \cup A_2 \mid B) = 0.2$
B. $P(A_1 \cup A_2 \mid B) = 0.5$
C. $P(A_1 \cup A_2 \mid B) = 0.8$
D. $P(A_1 \cup A_2 \mid B) = 0.9$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Consider a scenario where a ball is randomly placed in one of n boxes, each with probability P_i for box i. If the ball is in box i, a search of that box will uncover it with probability α_i. The conditional probability that the ball is in box j, given that a search of box i did not uncover it, is:

A. $P(\text{ ball in box } j \mid \text{ not found in box } i) = \frac{P_j}{1-\alpha_i P_i}$, if $j \neq i$

B. $P(\text{ ball in box } j \mid \text{ not found in box } i) = \frac{(1-\alpha_i)P_i}{1-\alpha_i P_i}$, if $j \neq i$

C. $P(\text{ ball in box } j \mid \text{ not found in box } i) = \frac{1-\alpha_i P_i}{P_j}$, if $j \neq i$

D. $P(\text{ ball in box } j \mid \text{ not found in box } i) = \frac{P_i}{1-\alpha_i}$, if $j \neq i$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

You have three coins in your pocket: two fair ones and one biased with probability p for heads and $1-p$ for tails.
Let F, U, and H denote the events that:

- F : The selected coin is fair.
- U : The selected coin is unfair.
- H : The coin lands heads up.

What is the probability that the selected coin is one of the fair coins, given that it lands heads up? (i.e., $P(F \mid H)$)

A. $\frac{1}{1+p}$

B. $\frac{p}{1+p}$

C. $\frac{1}{2}$

D. $\frac{2}{3}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Is it possible to have events A, B, C such that:

$$P(A \mid C) < P(B \mid C) \text{ and } P\left(A \mid C^c\right) < P\left(B \mid C^c\right), \text{ yet } P(A) > P(B) \text{ ?}$$

A. Yes, it is possible, but only if $P(C) = P\left(C^c\right)$.
B. Yes, it is always possible, irrespective of $P(C)$ being equal to $P\left(C^c\right)$.
C. No, it is not possible; such a scenario contradicts probability rules.
D. The answer cannot be determined from the information given.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Given probabilities:

$$\mathbb{P}(A) = \frac{1}{5}, \quad \mathbb{P}(B) = \frac{3}{10}, \quad \mathbb{P}(A \cup B) = \frac{11}{25}$$

and assuming events A and B are independent, what is $\mathbb{P}(A \cap B \mid A \cup B)$?

A. $\frac{3}{22}$
B. $\frac{11}{25}$
C. $\frac{1}{5}$
D. $\frac{2}{5}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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